

Team Notebook

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1 Data Structure

1.1 2d fenwick tree

```

struct BIT2D {
    vector<vector<int>>> t;
    int n, m;
    BIT2D(int n, int m) : n(n), m(m) {
        t.assign(n, vector<int>(m, 0));
    }
    void update(int x, int y, int add) {
        for (int i = x; i < n; i |= i + 1)
            for (int j = y; j < m; j |= j + 1) t[i][j] +=
add;
    }
    int sum(int x, int y) {
        int ans = 0;
        for (int i = x; i >= 0; i = (i & (i + 1)) - 1)
            for (int j = y; j >= 0; j = (j & (j + 1)) - 1)
ans += t[i][j];
        return ans;
    }
    int query(int lx, int rx, int ly, int ry) {
        return sum(rx, ry) - sum(lx - 1, ry) - sum(rx, ly -
1) + sum(lx - 1, ly - 1);
    }
};

```

1.2 centroid decomposition

```

struct Centroid_Decomposition{
    vector<int> adj[N];
    int sz[N], vis[N], aux;
    int pa[N];

    int find(int u, int p, int Sz){
        int mx = 0, c = 0; sz[u] = 1;
        for (auto v : adj[u]){
            if (v == p || vis[v]) continue;
            c ^= find(v, u, Sz);
            sz[u] += sz[v];
            mx = max(mx, sz[v]);
        }
        if (max(mx, Sz-sz[u]) <= Sz/2)
            aux = p, c = u;
        return c;
    }

    void build(int u, int p, int Sz){
        aux = -1, u = find(u, -1, Sz);
        if (aux != -1) sz[aux] = Sz-sz[u];
        vis[u] = 1, sz[u] = Sz, pa[u] = p;
        for (auto v : adj[u])
            if (!vis[v]) build(v, u, sz[v]);
    }
};

```

```

    }
} cd; // by NekoRolly

```

1.3 dynamic segment tree

```

struct nod {
    ll t;
    ll lazy;
    nod *l;
    nod *r;
};

static nod nods[ND];
static int id = 0;
nod *crearnodo() {
    nods[id].t = NEUT, nods[id].lazy = NEUT_LAZY, nods[id].l
= nods[id].r = NULL;
    return &nods[id++];
}

ll f(ll a, ll b) {
    return a + b;
}

void aply(ll val, nod *u, ll l, ll r) {
    u->t = (r - l + 1) * val;
    u->lazy = val;
}

struct segmentTree {
    ll n;
    nod *root = NULL;
    segmentTree(ll n) : n(n) {}
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) / 2;
        if (!u->l) u->l = crearnodo();
        if (!u->r) u->r = crearnodo();
        aply(u->lazy, u->l, l, m);
        aply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }

    ll query(ll ql, ll qr, nod *u, ll l, ll r) {
        if (qr < l || r < ql || !u) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
        ll m = (l + r) / 2;
        return f(query(ql, qr, u->l, l, m), query(ql, qr, u-
>r, m + 1, r));
    }

    ll query(ll ql, ll qr) {
        return query(ql, qr, root, 0, n - 1);
    }

    void update(ll ql, ll qr, ll val, nod *u, ll l, ll r) {
        if (qr < l || r < ql) return;
        if (!u) u = crearnodo();
        if (ql <= l && r <= qr) aply(val, u, l, r);
        else {
            push(u, l, r);

```

```

        ll m = (l + r) / 2;
        update(ql, qr, val, u->l, l, m);
        update(ql, qr, val, u->r, m + 1, r);
        u->t = f(u->l ? u->l->t : NEUT, u->r ? u->r->t :
NEUT);
    }
}

void update(ll ql, ll qr, ll val) {
    update(ql, qr, val, root, 0, n - 1);
}

};

```

1.4 fenwick tree

```

struct FenwickTree { // Index 1
    int n, ft[MX + 1];
    void update(int l, int v) {
        for (; l <= n; l += l & -l) ft[l] += val;
    }
    int query(int l) {
        int ans = 0;
        for (; l > 0; l -= l & -l) ans += t[l];
        return ans;
    }
    int query(int l, int r) { return query(r) - query(l -
1); }
};

```

1.5 hld

```

struct Graph {
    vector<vector<ll>>> adj;
    vector<ll> padre, prof, heavy, head, pos, base, L, R;
    segmentTree st;
    ll curpos, n;
    ll dfs(ll u, ll p) {
        padre[u] = p;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (ll v : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u);
                szu += szv;
                if (mxsz < szv) {
                    mxsz = szv;
                    heavy[u] = v;
                }
            }
        return szu;
    }
    void decompose(ll u, ll h, vector<ll> &a) {
        head[u] = h;
        L[u] = curpos;

```

```

    pos[u] = curpos++;
    base[pos[u]] = a[u - 1];
    if (heavy[u] != 0) decompose(heavy[u], h, a);
    for (ll v : adj[u])
        if (v != padre[u] && v != heavy[u]) decompose(v,
v, a);
    R[u] = curpos - 1;
}
void pupdate(ll u, ll val) {
    st.pupdate(pos[u], val);
}
void update(ll u, ll v, ll val) {
    while (head[u] != head[v]) {
        if (prof[head[u]] > prof[head[v]]) swap(u, v);
        st.update(pos[head[v]], pos[v], val);
        v = padre[head[v]];
    }
    if (prof[u] > prof[v]) swap(u, v);
    st.update(pos[u], pos[v], val);
}
ll query(ll u, ll v) {
    ll ans = INF;
    while (head[u] != head[v]) {
        if (prof[head[u]] > prof[head[v]]) swap(u, v);
        ans = f(ans, st.query(pos[head[v]], pos[v]));
        v = padre[head[v]];
    }
    if (prof[u] > prof[v]) swap(u, v);
    ans = f(ans, st.query(pos[u], pos[v]));
    return ans;
}
ll subtreequery(ll u) {
    return st.query(L[u], R[u]);
}
void subtreeupdate(ll u, ll val) {
    st.update(L[u], R[u], val);
}
}
Graph(vector<ll> &a, vector<pll> &aristas, ll root =
1) : n(a.size()) {
    adj.assign(n + 1, vector<ll>()), padre.assign(n + 1,
0), prof.assign(n + 1, 0);
    heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
    L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
    curpos = 0;
    for (auto [u, v] : aristas) {
        adj[u].pb(v);
        adj[v].pb(u);
    }
    prof[root] = 0;
    dfs(root, root);
    decompose(root, root, a);
    st = segmentTree(base);
}

```

```

    }
};

```

1.6 hld edge

```

struct Graph {
    vector<vector<pll>> adj;
    vector<ll> padre, wpadre, prof, heavy, head, pos, base,
L, R;
    segmentTree st;
    ll curpos;
    ll n;
    ll dfs(ll u, ll p, ll wp) {
        padre[u] = p;
        wpadre[u] = wp;
        ll szu = 1;
        ll mxsz = 0;
        heavy[u] = 0;
        for (auto [v, w] : adj[u])
            if (v != p) {
                prof[v] = prof[u] + 1;
                ll szv = dfs(v, u, w);
                szu += szv;
                if (mxsz < szv) mxsz = szv, heavy[u] = v;
            }
        return szu;
    }
    void decompose(ll u, ll h) {
        head[u] = h;
        L[u] = curpos;
        pos[u] = curpos++;
        base[pos[u]] = wpadre[u];
        if (heavy[u] != 0) decompose(heavy[u], h);
        for (auto [v, w] : adj[u])
            if (v != padre[u] && v != heavy[u]) decompose(v,
v);
        R[u] = curpos - 1;
    }
    void pupdate(ll u, ll v, ll val) {
        if (u == padre[v]) swap(u, v);
        st.pupdate(pos[u], val);
    }
    void update(ll u, ll v, ll val) {
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            st.update(pos[head[v]], pos[v], val);
            v = padre[head[v]];
        }
        if (prof[u] > prof[v]) swap(u, v);
        if (u != v) st.update(pos[u] + 1, pos[v], val);
    }
    ll query(ll u, ll v) {
        ll ans = INF;
        while (head[u] != head[v]) {
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
            if (prof[head[u]] > prof[head[v]]) swap(u, v);
        }
    }
}

```

```

    ans = f(ans, st.query(pos[head[v]], pos[v]));
    v = padre[head[v]];
}
if (prof[u] > prof[v]) swap(u, v);
if (u != v) ans = f(ans, st.query(pos[u] + 1,
pos[v]));
return ans;
}
ll subtreequery(ll u) {
    if (L[u] == R[u]) return 0;
    return st.query(L[u] + 1, R[u]);
}
void subtreeupdate(ll u, ll val) {
    if (L[u] == R[u]) return;
    st.update(L[u] + 1, R[u], val);
}
}
Graph(vector<pair<pll>, ll>> &aristas, ll root = 1) :
n(aristas.size() + 1) {
    adj.assign(n + 1, vector<pll>());
    padre.assign(n + 1, 0), wpadre.assign(n + 1, 0),
prof.assign(n + 1, 0);
    heavy.assign(n + 1, 0), head.assign(n + 1, 0),
pos.assign(n + 1, 0);
    L.assign(n + 1, 0), R.assign(n + 1, 0),
base.resize(n);
    curpos = 0;
    for (auto [edge, w] : aristas) {
        auto [u, v] = edge;
        adj[u].pb({v, w});
        adj[v].pb({u, w});
    }
    prof[root] = 0;
    dfs(root, root, 0LL);
    decompose(root, root);
    st = segmentTree(base);
}
};

```

1.7 merge sort tree

```

struct segmentTree {
    vector<vector<ll>> seg;
    ll n, N;
    segmentTree() {}
    segmentTree(vector<ll> &a) : n(a.size()) {
        N = 1;
        while (N < n) N <= 1;
        seg.assign(2 * N, vector<ll>());
        for (int i = 0; i < n; i++) seg[i + N] = {a[i]};
        for (int i = N - 1; i > 0; i--) {
            auto &l = seg[i < 1], &r = seg[i < 1 | 1];
            seg[i].resize((ll)l.size() + (ll)r.size());
            merge(all(l), all(r), seg[i].begin());
        }
    }
}

```

```
// cantidad de elementos mayores a "a" en el intervalo [ql,qr]
ll query1(ll ql, ll qr, ll a) {
    ll l = ql + N, r = qr + N;
    ll ans = 0;
    while (l <= r) {
        if (l & 1) {
            auto &v = seg[l++];
            ans += v.end() - upper_bound(all(v), a);
        }
        if (!(r & 1)) {
            auto &v = seg[r--];
            ans += v.end() - upper_bound(all(v), a);
        }
        l >>= 1;
        r >>= 1;
    }
    return ans;
}

// cantidad de elementos menores a "a" en el intervalo [ql,qr]
ll query2(ll ql, ll qr, ll a) {
    ll l = ql + N, r = qr + N;
    ll ans = 0;
    while (l <= r) {
        if (l & 1) {
            auto &v = seg[l++];
            ans += lower_bound(all(v), a) - v.begin();
        }
        if (!(r & 1)) {
            auto &v = seg[r--];
            ans += lower_bound(all(v), a) - v.begin();
        }
        l >>= 1;
        r >>= 1;
    }
    return ans;
}
};
```

1.8 multi ordered set

```
// find_by_order(k) -> return iterator to the k-th element (0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num - O(log n)

// using less_equal cause that lower_bound works as
upper_bound and viceversa
// also .find() does not work and you can only erase by
iterator now
// using for example erase(upper_bound(x))
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

```
template <typename T>
using ordered_set =
    tree<T, null_type, less_equal<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

ordered_set<int> s;
```

1.9 ordered set

```
// find_by_order(k) -> return iterator to the k-th element (0-indexed) - O(log n)
// order_of_key(num) -> # of items strictly smaller than num - O(log n)
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;
```

```
template <typename T>
using ordered_set =
    tree<T, null_type, less<T>, rb_tree_tag,
    tree_order_statistics_node_update>;

ordered_set<int> S;
```

1.10 persistent segment tree

```
const ll MX = 2e5 + 5;
const ll Q = 2e5 + 5;
const ll MXT = 2 * MX + 20 * Q;
struct PST {
    ll root[Q + 1], t[MXT], L[MXT], R[MXT];
    ll n, nodos, lastversion = 0;
    PST(vector<ll> &a) : n(a.size()) {
        nodos = 2 * n - 1;
        root[0] = 1;
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void build(vector<ll> &a, ll id, ll l, ll r) {
        if (l == r) t[id] = a[l];
        else {
            ll m = (l + r) / 2;
            ll idl = id + 1;
            ll idr = id + 2 * (m - l + 1);
            L[id] = idl;
            R[id] = idr;
            build(a, idl, l, m);
            build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(ll ql, ll qr, ll id, ll l, ll r) {
        if (ql <= l && r <= qr) return t[id];
        ll m = (l + r) / 2;
```

```
        if (qr <= m) return query(ql, qr, L[id], l, m);
        if (m + 1 <= ql) return query(ql, qr, R[id], m + 1, r);
        return f(query(ql, qr, L[id], l, m), query(ql, qr, R[id], m + 1, r));
    }
    ll query(ll ql, ll qr, ll version = -1) {
        if (version == -1) version = lastversion;
        return query(ql, qr, root[version], 0, n - 1);
    }
    void update(ll pos, ll val, ll prev, ll &id, ll l, ll r) {
        id = ++nodos;
        if (l == r) t[id] = val;
        else {
            ll m = (l + r) / 2;
            L[id] = L[prev];
            R[id] = R[prev];
            if (pos <= m) update(pos, val, L[prev], L[id], l, m);
            else update(pos, val, R[prev], R[id], m + 1, r);
            t[id] = f(t[L[id]], t[R[id]]);
        }
    }
    void update(ll pos, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        lastversion++;
        update(pos, val, root[version], root[lastversion], 0, n - 1);
    }
};
```

1.11 persistent segment tree lazy

```
const int MX = 2e5 + 5;
const int Q = 2e5 + 5;
const int LOG = 20;
const int ND = 2 * MX + 6 * Q * LOG;
const ll NEUT = 0;
const ll NEUT_LAZY = -2 * INF;
struct nod {
    ll t, lazy;
    nod *l, *r;
};
nod n[ND];
int id = 0;
nod *crearnodo(nod *u = NULL) {
    if (u) n[id].t = u->t, n[id].lazy = u->lazy,
    n[id].l = u->l, n[id].r = u->r;
    else n[id].t = NEUT, n[id].lazy = NEUT_LAZY,
    n[id].l = n[id].r = NULL;
    return &n[id++];
}
ll f(ll a, ll b) {
    return a + b;
```

```

}
void apply(ll val, nod *u, ll l, ll r) {
    u->t += (r - l + 1) * val;
    if (u->lazy == NEUT_LAZY) u->lazy = val;
    else u->lazy += val;
}
struct PST {
    nod *root[MX + 5];
    ll n;
    ll lastversion = -1;
    PST(vector<ll> &a) : n(a.size()) {
        root[++lastversion] = crearnodo();
        build(a, root[0], 0, n - 1);
    }
    void build(vector<ll> &a, nod *u, ll l, ll r) {
        if (!u) u = crearnodo();
        if (l == r) u->t = a[l];
        else {
            ll m = (l + r) >> 1;
            build(a, u->l, l, m);
            build(a, u->r, m + 1, r);
            u->t = f(u->l->t, u->r->t);
        }
    }
    void push(nod *u, ll l, ll r) {
        if (u->lazy == NEUT_LAZY || l == r) return;
        ll m = (l + r) >> 1;
        u->l = crearnodo(u->l), u->r = crearnodo(u->r);
        apply(u->lazy, u->l, l, m);
        apply(u->lazy, u->r, m + 1, r);
        u->lazy = NEUT_LAZY;
    }
    nod *update(ll ql, ll qr, ll val, nod *prev, ll l, ll r) {
        if (qr < l || r < ql) return prev;
        nod *u = crearnodo(prev);
        if (ql <= l && r <= qr) apply(val, u, l, r);
        else {
            push(u, l, r);
            ll m = (l + r) >> 1;
            u->l = update(ql, qr, val, u->l, l, m);
            u->r = update(ql, qr, val, u->r, m + 1, r);
            u->t = f(u->l->t, u->r->t);
        }
        return u;
    }
    void update(ll ql, ll qr, ll val, ll version = -1) {
        if (version == -1) version = lastversion;
        root[++lastversion] = update(ql, qr, val,
            root[version], 0, n - 1);
    }
    ll query(ll ql, ll qr, nod *u, ll l, ll r) {
        if (!u || qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return u->t;
        push(u, l, r);
    }
}

```

```

    ll m = (l + r) >> 1;
    return f(query(ql, qr, u->l, l, m), query(ql, qr, u->r, m + 1, r));
}
ll query(ll ql, ll qr, ll version = -1) {
    if (version == -1) version = lastversion;
    return query(ql, qr, root[version], 0, n - 1);
}
};

```

1.12 segment tree

```

struct IterativeSegmentTree { // Index 0
    int n;
    ll ST[2 * MX];
    const ll NEUT = 0;
    void build(vector<ll> &a) {
        n = a.size();
        for (int i = n; i < (n << 1); i++) ST[i] = a[i - n];
        for (int i = n - 1; i > 0; i--) ST[i] = f(ST[i << 1], ST[i << 1 | 1]);
    }
    ll f(ll x, ll y) { return max(x, y); }
    void update(int i, ll val) {
        for (ST[i += n] = val; i >= 1; i >>= 1) ST[i] = f(ST[i << 1], ST[i << 1 | 1]);
    }
    ll query(int l, int r) { // [l, r)
        ll ansL = NEUT, ansR = NEUT;
        for (l += n, r += n; l < r; l >= 1, r >= 1) {
            if (l & 1) ansL = f(ansL, ST[l++]);
            if (r & 1) ansR = f(ST[--r], ansR);
        }
        return f(ansL, ansR);
    }
};

```

1.13 segment tree 2d

```

struct ST2D { //Index 0
    int n, m;
    vector<vector<ll>> t;
    ll f(ll a, ll b) {
        return a + b;
    }
    ST2D(vector<vector<ll>> &a) {
        n = a.size();
        m = a[0].size();
        t.assign(2 * n, vector<ll>(2 * m, 0));
        for (int i = 0; i < n; i++)
            for (int j = 0; j < m; j++) t[i + n][j + m] = a[i][j];
        for (int i = n; i < 2 * n; i++)
            for (int j = m - 1; j > 0; j--) t[i][j] = f(t[i], t[i << 1], t[i << 1 | 1]);
    }
};

```

```

    for (int i = n - 1; i > 0; i--)
        for (int j = 1; j < 2 * m; j++) t[i][j] = f(t[i << 1][j], t[i << 1 | 1][j]);
}
void update(int x, int y, ll val) {
    x += n, y += m, t[x][y] = val;
    for (int j = y; j > 1; j >= 1) t[x][j >> 1] = f(t[x][j], t[x][j ^ 1]);
    for (int i = x >> 1; i > 0; i >= 1) {
        int j = y;
        t[i][j] = f(t[i << 1][j], t[i << 1 | 1][j]);
        for (; j > 1; j >= 1) t[i][j >> 1] = f(t[i][j], t[i][j ^ 1]);
    }
}
ll queryY(int i, int l, int r) {
    ll ans = 0;
    for (l += m, r += m + 1; l < r; l >= 1, r >= 1) {
        if (l & 1) ans = f(ans, t[l][l++]);
        if (r & 1) ans = f(t[r][--r], ans);
    }
    return ans;
}
ll query(int lx, int rx, int ly, int ry) {
    ll ans = 0;
    for (x1 += n, x2 += n + 1; x1 < x2; x1 >= 1, x2 >= 1) {
        if (x1 & 1) ans = f(ans, queryY(x1++, y1, y2));
        if (x2 & 1) ans = f(queryY(--x2, y1, y2), ans);
    }
    return ans;
}
};

```

1.14 segment tree beats

```

struct node {
    ll sum;
    ll max1, max2, cont; // min para chmax
};
node f(node a, node b) { // simetrico con min para chmax
    node ans;
    ans.sum = a.sum + b.sum;
    ans.max1 = max(a.max1, b.max1);
    if (a.max1 == b.max1) ans.max2 = max(a.max2, b.max2),
    ans.cont = a.cont + b.cont;
    else if (a.max1 > b.max1) ans.max2 = max(a.max2, b.max1), ans.cont = a.cont;
    else ans.max2 = max(a.max1, b.max2), ans.cont = b.cont;
    return ans;
}
struct SegmentTree {
    vector<node> t;
    int n;
    SegmentTree(vector<ll> &a) : n(a.size()) {

```

```

t.resize(2 * n);
build(a, 1, 0, n - 1);
}
void build(vector<ll> &a, int id, int l, int r) {
    if (l == r) t[id] = {a[l], a[l], -INF, 1}; // chmax:
-INF -> INF
    else {
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        build(a, idl, l, m), build(a, idr, m + 1, r);
        t[id] = f(t[idl], t[idr]);
    }
}
void apply(ll val, int id) {
    if (t[id].max1 <= val) return; // chmax: (<=) ->
(>=)
    t[id].sum += t[id].cont * (val - t[id].max1);
    t[id].max1 = val;
}
void push(int id, int l, int r) {
    if (l == r) return;
    int m = (l + r) >> 1;
    int idl = id + 1;
    int idr = id + 2 * (m - l + 1);
    apply(t[id].max1, idl);
    apply(t[id].max1, idr);
}
void update(int ql, int qr, ll val, int id, int l, int
r) {
    if (qr < l || r < ql || t[id].max1 <= val)
return; //chmax: (<=) -> (>=)
    if (ql <= l && r <= qr && t[id].max2 < val)
apply(val, id); // chmax: (<) -> (>)
    else {
        push(id, l, r);
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        update(ql, qr, val, idl, l, m), update(ql, qr,
val, idr, m + 1, r);
        t[id] = f(t[idl], t[idr]);
    }
}
ll query(int ql, int qr, int id, int l, int r) {
    if (qr < l || r < ql) return NEUT;
    if (ql <= l && r <= qr) return t[id].sum;
    push(id, l, r);
    int m = (l + r) >> 1;
    int idl = id + 1;
    int idr = id + 2 * (m - l + 1);
    return query(ql, qr, idl, l, m) + query(ql, qr, idr,
m + 1, r);
}
};

```

1.15 segment tree lazy

```

struct SegmentTree { // id: index 1. Vector a: Index 0
    vector<ll> t, lazy;
    int n;
    SegmentTree(vector<ll> &a) : n(a.size()) {
        t.assign(2 * n, NEUT), lazy.assign(2 * n,
NEUT_LAZY);
        build(a, 1, 0, n - 1);
    }
    ll f(ll a, ll b) {
        return a + b;
    }
    void apply(ll val, int id, int l, int r) {
        t[id] += (r - l + 1) * val;
        lazy[id] += val;
    }
    void build(vector<ll> &a, int id, int l, int r) {
        if (l == r) t[id] = a[l];
        else {
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            build(a, idl, l, m), build(a, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    void push(int id, int l, int r) {
        if (l == r || lazy[id] == NEUT_LAZY) return;
        int m = (l + r) >> 1;
        int idl = id + 1;
        int idr = id + 2 * (m - l + 1);
        apply(lazy[id], idl, l, m);
        apply(lazy[id], idr, m + 1, r);
        lazy[id] = NEUT_LAZY;
    }
    void update(int ql, int qr, ll val, int id, int l, int
r) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) apply(val, id, l, r);
        else {
            push(id, l, r);
            int m = (l + r) >> 1;
            int idl = id + 1;
            int idr = id + 2 * (m - l + 1);
            update(ql, qr, val, idl, l, m), update(ql, qr,
val, idr, m + 1, r);
            t[id] = f(t[idl], t[idr]);
        }
    }
    ll query(int ql, int qr, int id, int l, int r) {
        if (qr < l || r < ql) return NEUT;
        if (ql <= l && r <= qr) return t[id];
        push(id, l, r);
        int m = (l + r) >> 1;
    }
};

```

```

int idl = id + 1;
int idr = id + 2 * (m - l + 1);
return f(query(ql, qr, idl, l, m), query(ql, qr,
idr, m + 1, r));
}
};

```

1.16 sparse table

```

const int LG = __lg(MX) + 1;
int st[MX][LG];
void build(vector<int> &A) { // O(|f| n log n)
    int n = A.size();
    for (int i = 0; i < n; i++) st[i][0] = A[i];
    for (int k = 1; k < LG; k++) {
        for (int i = 0; i + (1 << k) <= n; i++) {
            st[i][k] = min(st[i][k - 1], st[i + (1 << (k -
1))][k - 1]);
        }
    }
}
int query(int l, int r) { // O(|f|)
    int lg = __lg(r - l + 1);
    return min(st[l][lg], st[r - (1 << lg) + 1][lg]);
}

```

1.17 trie bitmask

```

struct Trie{
    int t[N*B][2], sz[N*B];
    int cur;

    void insert(int x){
        int u = 0; sz[u]++;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k] == 0) t[u][k] = ++cur;
            u = t[u][k], sz[u]++;
        }
    }

    void erase(int x){
        int u = 0; sz[u]--;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            u = t[u][k], sz[u]--;
        }
    }

    int get_min(int x){ // min xor
        int u = 0, ans = 0;
        for (int i=B-1; i>=0; i--){
            int k = x>>i&1;
            if (t[u][k] != 0 && sz[t[u][k]] > 0) u = t[u]
[k];
        }
    }
};

```

```

        else u = t[u][k^1], ans ^= 1<<i;
    }
    return ans;
}

int get_max(int x){ // max xor
    int u = 0, ans = 0;
    for (int i=B-1; i>=0; i--){
        int k = x>>i&1;
        if (t[u][k^1] != 0 && sz[t[u][k^1]] > 0)
            u = t[u][k^1], ans ^= 1<<i;
        else u = t[u][k];
    }
    return ans;
}
} tr; // by NekoRolly

```

1.18 trie string

```

struct Trie{
    int t[N][K], sz[N], cur;

    void insert(string &s){
        int u = 0; sz[u]++;
        for (char &c : s){
            int k = c-'a';
            if (t[u][k] == 0) t[u][k] = ++cur;
            u = t[u][k], sz[u]++;
        }
    }

    void erase(string &s){
        int u = 0; sz[u]--;
        for (char &c : s){
            int k = c-'a';
            u = t[u][k], sz[u]--;
        }
    }
} tr; // by NekoRolly

```

2 Geometry

2.1 convex hull trick

```

struct Line{
    ll m, b;

    ll y(ll x){ return m*x+b; }
};

struct Convex_Hull_Track{
    vector<Line> vec;
    int l = 0;

```

```

    bool check(int i){
        double x1 = double(vec[i-1].b - vec[i+1].b)/
            (vec[i+1].m - vec[i-1].m);
        double x2 = double(vec[i].b - vec[i+1].b)/
            (vec[i+1].m - vec[i].m);
        return x1 >= x2;
    }

    void add(Line L){
        vec.push_back(L);
        while (vec.size() >= 3 && check(vec.size()-2))
            vec.erase(vec.end()-2);
        l = min(l, int(vec.size()-1));
    }

    ll query(ll x){ // O(1) amortizado
        while (l+1 < vec.size() && vec[l].y(x) >
            vec[l+1].y(x)) l++;
        return vec[l].y(x);
    }

    ll find(ll x){ // O(logn)
        int l = 0, r = vec.size();
        while (l+1 < r){
            int m = (l+r)>>1;
            if (vec[m-1].y(x) > vec[m].y(x)) l = m;
            else r = m;
        }
        return vec[l].y(x);
    }
} cht; // by NekoRolly

```

2.2 li chao tree

```

struct Line{
    ll m, b;

    ll y(ll x){ return m*x+b; }
};

struct Li_Chao_Tree{
    Line t[N*4];

    void update(int id, int l, int r, Line L){
        if (l+1 == r || t[id].b == -infll){
            if (L.y(l) > t[id].y(l)) t[id] = L;
            return;
        }
        int m = (l+r)>>1;
        if (t[id].y(m) < L.y(m)) swap(t[id], L);
        if (t[id].m > L.m) update(id<<1, l, m, L);
        else update(id<<1|1, m, r, L);
    }

    ll query(int id, int l, int r, ll x){

```

```

        if (l+1 == r) return t[id].y(x);
        int m = (l+r)>>1;
        if (x < m) return max(t[id].y(x), query(id<<1, l, m,
            x));
        else return max(t[id].y(x), query(id<<1|1, m, r,
            x));
    }
} lct; // by NekoRolly

```

3 Graphs

3.1 2sat

```

struct SAT { // Index_1
    vector<vector<int>> adj, adjt;
    vector<bool> used;
    vector<int> comp;
    int n;
    stack<int> st;
    SAT(int n) : n(n) {
        adj.resize(2 * n + 1), adjt.resize(2 * n + 1);
        used.assign(2 * n + 1, 0), comp.assign(2 * n + 1,
            0);
    }

    void add(int p, bool bp, int q, bool bq) { // add p -> q
        p += (!bp) * n;
        q += (!bq) * n;
        adj[p].pb(q);
        adjt[q].pb(p);
    }

    void addor(int p, bool bp, int q, bool bq) { // add p or
        q
        add(p, !bp, q, bq);
        add(q, !bq, p, bp);
    }

    void addxor(int p, bool bp, int q, bool bq) { // add p
        xor q
        addor(p, bp, q, bq);
        addor(p, !bp, q, !bq);
    }

    void addnxor(int p, bool bp, int q, bool bq) { // add
        ~(p xor q)
        addor(p, !bp, q, bq);
        addor(p, bp, q, !bq);
    }

    void dfs(int u) {
        used[u] = 1;
        for (int v : adj[u])
            if (!used[v]) dfs(v);
        st.push(u);
    }

    void _dfs(int u, int c) {
        comp[u] = c;
        for (int v : adjt[u])

```



```

        if (!comp[v]) _dfs(v, c);
    }
    bool solve(vector<bool> &ans) {
        ans.resize(n + 1);
        for (int u = 1; u <= 2 * n; u++)
            if (!used[u]) dfs(u);
        int c = 0;
        while (!st.empty()) {
            int u = st.top();
            st.pop();
            if (!comp[u]) _dfs(u, ++c);
        }
        for (int p = 1; p <= n; p++) {
            if (comp[p] == comp[p + n]) return false;
            ans[p] = comp[p] > comp[p + n];
        }
        return true;
    }
};

```

3.2 bellman ford

```

// find distance from source node to all nodes.
// supports negative edge weights.
// returns true if a negative cycle is detected.
bool bellmanFord(int s, int n) { // O(VE)
    d.assign(n + 1, INF);
    vector<bool> cycle(n, false);
    bool f = false;
    for (int i = 0; i < n; i++) {
        f = false;
        for (int u = 0; u < n; u++) {
            for (auto [v, w] : adj[u]) {
                if (d[u] != INF && d[u] + w < d[v]) {
                    d[v] = d[u] + w;
                    parent[v] = u;
                    f = true;
                    if (i == n - 1) cycle[v] = true;
                }
            }
        }
        if (!f) return false;
    }
    return true;
}

```

3.3 blossom

```

struct Graph {
    vector<int> adj[MX];
    int p[MX];
    int match[MX];
    // match[u] is the node match with u
    // or -1 if it does not belong to the matching

```

```

    int base[MX]; // base of the flower
    bool used[MX];
    bool blossom[MX];

    void addEdge(int u, int v);

    int lca(int u, int v, int n) { // O(n)
        vector<bool> marked(n);
        while (true) {
            u = base[u]; // blossom compression
            marked[u] = true;
            if (match[u] == -1) break; // we reach the
            root : unexposed vertex
            u = p[match[u]];
        }
        while (true) {
            v = base[v];
            if (marked[v]) return v;
            v = p[match[v]];
        }
        return -1;
    }

    void markPath(int u, int b, int children) {
        while (base[u] != b) {
            blossom[base[u]] = blossom[base[match[u]]] =
true;
            p[u] = children, children = match[u], u =
p[match[u]];
        }
    }

    void compress(int u, int v, int n, queue<int> &q) {
        int curBase = lca(u, v, n); // base of the flower
        assert(curBase != -1);
        fill(blossom, blossom + n, false);

        markPath(u, curBase, v);
        markPath(v, curBase, u);
        for (int i = 0; i < n; i++) {
            if (blossom[base[i]]) {
                base[i] = curBase;
                if (!used[i]) used[i] = true, q.push(i);
            }
        }
    }

    int findPath(int root, int n) {
        for (int i = 0; i < n; i++) base[i] = i, used[i] =
false, p[i] = -1;
        used[root] = true;
        queue<int> q;
        q.push(root);
        while (!q.empty()) {
            int u = q.front();

```

```

        }
        while (!q.empty()) {
            int u = q.front();

```

```

        q.pop();
        for (int to : adj[u]) {
            if (base[u] == base[to] || match[u] == to)
continue;
            if (to == root || (match[to] != -1 &&
p[match[to]] != -1))
                compress(u, to, n, q); // we find a
blossom
            else if (p[to] == -1) {
                // augmenting path or continue bfs
                p[to] = u;
                if (match[to] == -1) return to; // we
find an augmenting path
                // we continue the bfs
                to = match[to], used[to] = true,
q.push(to);
            }
        }
        return -1;
    }

    int maxMatching(int n) { // O(n^3)
        fill(match, match + n, -1);
        int ans = 0;
        for (int u = 0; u < n; u++) {
            if (match[u] == -1) {
                int v = findPath(u, n);
                if (v != -1) {
                    ans++;
                    while (v != -1) {
                        int nextVertex = match[p[v]];
                        match[v] = p[v], match[p[v]] = v, v
= nextVertex;
                    }
                }
                return ans;
            }
        }
    }
};

```

3.4 dinic

```

// This implementation is for small graphs G(V, E) (|V| <=
3000 aprox)
// where O(|V|^2) memory will fit the memory limit.
using Cap = long long;
const int MX = 3000;
const Cap INF = 1e18;
struct Graph {
    vector<int> adj[MX];
    Cap cap[MX][MX];
    Cap flow[MX][MX];
    bool added[MX][MX];

```



```

int level[MX];
int nextEdge[MX];

void clear(int n) {
    for (int i = 0; i < n; i++) {
        adj[i].clear();
        for (int j = 0; j < n; j++) cap[i][j] = 0,
        flow[i][j] = 0, added[i][j] = false;
    }
}

void addEdge(int u, int v, Cap w, bool dir) {
    if (!added[u][v]) {
        adj[u].push_back(v);
        adj[v].push_back(u);
        added[u][v] = added[v][u] = true;
    }
    cap[u][v] += w;
    if (!dir) cap[v][u] += w;
}

Cap dfs(int u, int t, Cap f) { // O(E)
    if (u == t) return f;
    for (int &i = nextEdge[u]; i < adj[u].size(); i++) {
        int v = adj[u][i];
        Cap cf = cap[u][v] - flow[u][v];
        if (cf == 0 || level[v] != level[u] + 1)
            continue;
        Cap ret = dfs(v, t, min(f, cf));
        if (ret > 0) {
            flow[u][v] += ret;
            flow[v][u] -= ret;
            return ret;
        }
    }
    return 0;
}

bool bfs(int s, int t, int n) { // O(E)
    fill(level, level + n, -1);
    queue<int> q({s});
    level[s] = 0;
    while (!q.empty()) {
        int u = q.front();
        nextEdge[u] = 0;
        q.pop();
        for (int v : adj[u]) {
            Cap cf = cap[u][v] - flow[u][v];
            if (level[v] == -1 && cf > 0) {
                level[v] = level[u] + 1;
                q.push(v);
            }
        }
    }
    return level[t] != -1;
}

```

```

}

Cap maxFlow(int s, int t, int n) {
    // General: O(E * V^2), O(E * V + V * |F|)
    // Unit Cap: O(E * min(E^(1/2), V^(2/3)))
    // Unit Network: O(E * V^(1/2))
    // In unit network, all the edges have unit capacity
    // and for any vertex except s and t either the
    // incoming or outgoing edge is unique.
    Cap ans = 0;
    while (bfs(s, t, n)) { // O(V) iterations
        // after bfs, the graph is a DAG
        while (Cap inc = dfs(s, t, INF)) ans += inc;
    }
    return ans;
}
} G;

```

3.5 dominator tree

```

//idom[i]=parent of i in dominator tree with root=rt, or -1
//if not exists
int
n, rnk[MXN], pre[MXN], anc[MXN], idom[MXN], semi[MXN], low[MXN];
vector<int> g[MXN], rev[MXN], dom[MXN], ord;
void dfspre(int pos) {
    rnk[pos] = SZ(ord); ord.pb(pos);
    for (auto x : g[pos]) {
        rev[x].pb(pos);
        if (rnk[x] == n) pre[x] = pos, dfspre(x);
    }
}
int eval(int v) {
    if (anc[v] < n && anc[anc[v]] < n) {
        int x = eval(anc[v]);
        if (rnk[semi[low[v]]] > rnk[semi[x]]) low[v] = x;
        anc[v] = anc[anc[v]];
    }
    return low[v];
}
void dominators(int rt) {
    for (int i = 0; i < n; i++) {
        dom[i].clear(); rev[i].clear();
        rnk[i] = pre[i] = anc[i] = idom[i] = n;
        semi[i] = low[i] = i;
    }
    ord.clear(); dfspre(rt);
    for (int i = SZ(ord) - 1; i >= 0; i--) {
        int w = ord[i];
        for (int v : rev[w]) {
            int u = eval(v);
            if (rnk[semi[w]] > rnk[semi[u]]) semi[w] = semi[u];
        }
        dom[semi[w]].pb(w); anc[w] = pre[w];
        for (int v : dom[pre[w]]) {

```

```

            int u = eval(v);
            idom[v] = (rnk[pre[w]] > rnk[semi[u]] ? u : pre[w]);
        }
        dom[pre[w]].clear();
    }
    for (int w : ord) if (w != rt && idom[w] != semi[w])
        idom[w] = idom[idom[w]];
    for (int i = 0; i < n; i++) if (idom[i] == n) idom[i] = -1;
}

```

3.6 dsu on tree

```

vector<int> adj[N];
int L[N], V[N], sz[N], cur;

void dfs1(int u, int p) {
    L[u] = cur, V[cur++] = u, sz[u] = 1;
    for (int v : adj[u]) {
        if (v == p) continue;
        dfs1(v, u);
        sz[u] += sz[v];
    }
}

void dfs2(int u, int p, bool flag) {
    int big = -1;
    for (int v : adj[u]) {
        if (v != p && (big == -1 || sz[big] < sz[v])) big = v;
    }
    for (int v : adj[u]) {
        if (v != p && v != big) dfs2(v, u, true);
        if (big != -1) dfs2(big, u, false);
        update(a[u], 1); // add
        for (int v : adj[u]) {
            if (v == p || v == big) continue;
            for (int i = L[v]; i < L[v] + sz[v]; i++)
                update(a[V[i]], 1); // add
        }
        ans[u] = dif;
        if (!flag) return;
        for (int i = L[u]; i < L[u] + sz[u]; i++)
            update(a[V[i]], -1); // remove
    } // by NekoRolly
}

```

3.7 floyd warshall

```

void Ini(int n) {
    for (int u = 0; u < n; u++) {
        for (int v = 0; v < n; v++) d[u][v] = INF, parent[u][v] = -1;
        d[u][u] = 0;
    }
}

void addEdge(int u, int v, int w) {
    if (w < d[u][v]) {

```

```

        d[u][v] = w;
        parent[u][v] = u;
    }
}

void floydWarshall(int n) { // O(V^3)
    for0(k, n) for0(u, n) for0(v, n) {
        if (d[u][k] == INF) continue;
        if (d[k][v] == INF) continue;
        if (d[u][k] + d[k][v] < d[u][v]) {
            d[u][v] = max(-INF, d[u][k] + d[k][v]);
            parent[u][v] = parent[k][v];
        }
    }

    // negative cycles
    for0(u, n) for0(v, n) for0(k, n) {
        if (d[k][k] < 0 && d[u][k] != INF && d[k][v] != INF) {
            d[u][v] = -INF;
        }
    }
}

```

3.8 kuhn

```

// Maximum matching for bipartite unweighted graphs
struct Graph {
    int match[MX];
    bool vis[MX];
    vector<int> adj[MX];
    void addEdge(int u, int v);

    bool dfs(int u) {
        if (vis[u]) return false;
        vis[u] = true;
        for (int v : adj[u]) {
            if (match[v] == -1 || dfs(match[v])) {
                match[v] = u, match[u] = v;
                return true;
            }
        }
        return false;
    }

    int maximumMatching(int n) { // O(V * E)
        // If the bipartition is (V1, V2) we can iterate
        // from the side
        // that have less nodes (say is V1) and achieve O(V1
        // * E)
        fill(match, match + n, -1);
        int ans = 0;
        for (int u = 0; u < n; u++) {
            if (match[u] == -1) {
                fill(vis, vis + n, false);
            }
        }
    }
}

```

```

        ans += dfs(u);
    }
    return ans;
}
} G;

```

3.9 lca

```

int tIn[MX], tOut[MX];
int anc[MX][LG];
int timer;

void dfs(int u) {
    tIn[u] = timer++;
    for (int j = 1; j < LG; j++) anc[u][j] = anc[anc[u][j - 1]][j - 1];
    for (int v : adj[u]) {
        if (v == anc[u][0]) continue;
        anc[v][0] = u;
        dfs(v);
    }
    tOut[u] = timer++;
}

bool isAnc(int u, int v) { // is u ancestor of v ?
    return tIn[u] <= tIn[v] && tOut[u] >= tOut[v];
}

int lca(int u, int v) { // O(log n)
    if (isAnc(u, v)) return u;
    if (isAnc(v, u)) return v;
    for (int i = LG - 1; i >= 0; i--) {
        if (!isAnc(anc[u][i], v)) u = anc[u][i];
    }
    return anc[u][0];
}

```

3.10 maxflow mincost

```

using Cap = long long;
using Cost = long long;

const int MX = 5000;
const Cap INF_CAP = 1e18;
const Cost INF_COST = 1e18;

struct Edge {
    int to;
    Cap flow, cap;
    Cost cost;
    int rev; // index of the backward edge in the adj list
    of to
    Edge(int to, Cap cap, Cost cost, int rev)
        : to(to), flow(0), cap(cap), cost(cost), rev(rev) {}
}

```

```

};

struct Graph {
    vector<Edge> adj[MX];
    int parentEdge[MX];
    Cost pot[MX];
    //|pot[u]| <= the maximum sum of absolute cost in a path
    // which is also bounded by [(V - 1) * C]
    // where C is the maximum value of |cost(e)| for all
    edges e
    void clear(int n) {
        for (int i = 0; i < n; i++) {
            adj[i].clear();
            pot[i] = 0;
        }
    }

    void addEdge(int u, int v, Cap w, Cost cost, bool dir) {
        adj[u].push_back(Edge(v, w, cost, adj[v].size()));
        adj[v].push_back(Edge(u, 0, -cost, adj[u].size() - 1));
        if (!dir) addEdge(v, u, w, cost, true);
    }

    void spfa(int s, int n) { // O(E V)
        vector<bool> inQueue(n, false);
        for (int i = 0; i < n; i++) pot[i] = INF_COST;
        queue<int> q;
        pot[s] = 0;
        inQueue[s] = true;
        q.push(s);
        while (!q.empty()) {
            int u = q.front();
            q.pop();
            inQueue[u] = false;
            for (Edge e : adj[u]) {
                int v = e.to;
                if (e.cap - e.flow > 0 && pot[u] + e.cost <
                    pot[v]) {
                    pot[v] = pot[u] + e.cost;
                    if (!inQueue[v]) q.push(v);
                    inQueue[v] = true;
                }
            }
        }
    }

    void pushFlow(int s, int t, Cap inc) {
        int v = t;
        while (v != s) {
            Edge &backward = adj[v][parentEdge[v]];
            int u = backward.to;
            Edge &forward = adj[u][backward.rev];
            forward.flow += inc;
            backward.flow -= inc;
        }
    }
}

```

```

    v = u;
}
}

pair<Cap, Cost> dijkstra(int s, int t, int n) { // O(E log V)
    //<flow, cost>
    using Path = pair<Cost, int>; //<weight, node>
    priority_queue<Path, vector<Path>, greater<Path>> q;
    vector<Cost> d(n, INF_COST);
    vector<Cap> residualCap(n, 0);
    d[s] = 0;
    residualCap[s] = INF_CAP;
    q.push(Path(d[s], s));
    while (!q.empty()) {
        auto [dist, u] = q.top();
        q.pop();
        if (dist != d[u]) continue;
        for (Edge e : adj[u]) {
            int v = e.to;
            Cap cf = e.cap - e.flow;
            Cost cost = e.cost + pot[u] - pot[v];
            if (cf > 0 && d[u] + cost < d[v]) {
                d[v] = d[u] + cost;
                q.push(Path(d[v], v));
                residualCap[v] = min(residualCap[v],
                    parentEdge[v] = e.rev;
            }
        }
    }
    if (d[t] == INF_COST) return {0, 0};
    for (int i = 0; i < n; i++) {
        if (pot[i] != INF_COST) pot[i] += d[i];
    }
    Cap cf = residualCap[t];
    pushFlow(s, t, cf);
    return {cf, pot[t] * cf};
}

pair<Cap, Cost> minCostFlow(int s, int t, int n) {
    // O(E log V * maxFlow)
    // maxFlow <= V * U, where U is the maximum capacity
    // Assumption: Initially no negative cycles
    //<maxFlow, minCost>
    spfa(s, n); // not necessary if there is no negative
edges
    pair<Cap, Cost> inc;
    Cap flow = 0;
    Cost cost = 0;
    do {
        inc = dijkstra(s, t, n);
        flow += inc.first;
        cost += inc.second;
    } while (inc.first > 0);
}

```

```

    return {flow, cost};
}
} G;

```

3.11 tree bridge

```

vector<int> adj[MX];
int tin[MX], low[MX], timer;
vector<pii> bridges;
int component[MX];
vector<int> nods[MX];
vector<array<int, 3>> adjcomp[MX];
void dfs(int u, int p) {
    tin[u] = low[u] = ++timer;
    for (int v : adj[u])
        if (v != p) {
            if (tin[v]) low[u] = min(low[u], tin[v]);
            else dfs(v, u), low[u] = min(low[u], low[v]);
            if (tin[u] < low[v]) bridges.pb({u, v});
        }
}
bool isbridge(int u, int v) {
    if (tin[u] > tin[v]) swap(u, v);
    return tin[u] < low[v];
}
void dfsc(int u, int comp) {
    component[u] = comp;
    nods[comp].pb(u);
    for (int v : adj[u])
        if (!isbridge(u, v) && !component[v]) dfsc(v, comp);
}
int tree_bridge(int n) {
    dfs(1, 1);
    int comp = 0;
    for (int u = 1; u <= n; u++)
        if (!component[u]) dfsc(u, ++comp);
    for (auto [u, v] : bridges) {
        int cu = component[u], cv = component[v];
        adjcomp[cu].pb({cv, u, v});
        adjcomp[cv].pb({cu, u, v});
    }
    return comp;
}

```

4 Math

4.1 cht

```

struct Line {
    mutable ll a, b, c;

    bool operator<(Line r) const {
        return a < r.a;
    }
}

```

```

bool operator<(ll x) const {
    return c < x;
}

};
// dynamically insert `a*x + b` lines and query for maximum
// at any x all operations have complexity O(log N)
struct LineContainer : multiset<Line, less<>> {
    ll div(ll a, ll b) {
        return a / b - ((a ^ b) < 0 && a % b);
    }

    bool isect(iterator x, iterator y) {
        if (y == end()) return x->c = INF, 0;
        if (x->a == y->a) x->c = x->b > y->b ? INF : -INF;
        else x->c = div(y->b - x->b, x->a - y->a);
        return x->c >= y->c;
    }

    void add(ll a, ll b) {
        // a *= -1, b *= -1 // for min
        auto z = insert({a, b, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->c >= y->c)
            isect(x, erase(y));
    }

    ll query(ll x) {
        if (empty()) return -INF; // INF for min
        auto l = *lower_bound(x);
        return l.a * x + l.b;
        // return -l.a * x - l.b; // for min
    }
};

```

4.2 criba

```

bool criba[MX];
ll primerfactor[MX], phi[MX];
// criba para primerfactor, phi O(nlog(log(n)))
void completar() {
    for (int i = 0; i < MX; i++) {
        criba[i] = 1;
        primerfactor[i] = i;
        phi[i] = i;
    }
    criba[0] = 0;
    criba[1] = 0;
    for (ll i = 2; i < MX; i++) {
        if (criba[i]) {
            phi[i]--;
            for (ll j = 2 * i; j < MX; j += i) {
                criba[j] = 0;
                phi[j] -= phi[j] / i;
            }
        }
    }
}

```

```

        if (primerfactor[j] == j) primerfactor[j] =
i;
    }
}
}
vector<pll> factorizacioncriba(ll n) { // O(log(n))
    vector<pll> ans;
    while (n > 1) {
        ll f = primerfactor[n];
        ll pot = 0;
        while (primerfactor[n] == f) {
            n /= f;
            pot++;
        }
        ans.pb({f, pot});
    }
    return ans;
}
vector<ll> divisorescriba(ll n) { // O(d(x))
    vector<ll> fact = {1};
    while (n > 1) {
        ll f = primerfactor[n];
        ll sz = fact.size();
        ll divisor = 1;
        while (n % f == 0) {
            divisor *= f;
            n /= f;
            rep(i, sz) fact.pb(fact[i] * divisor);
        }
        return fact;
    }
}

```

4.3 diophantine

```

ll extended_gcd(ll a, ll b, ll &x, ll &y) {
    x = 1, y = 0;
    ll x1 = 0, y1 = 1;
    while (b) {
        ll q = a / b;
        tie(x, x1) = make_pair(x1, x - q * x1);
        tie(y, y1) = make_pair(y1, y - q * y1);
        tie(a, b) = make_pair(b, a - q * b);
    }
    return a < 0 ? (x *= -1, y *= -1, a *= -1, a) : a;
}
bool diophantine(ll a, ll b, ll c, ll &x, ll &y, ll &g) {
    if (a == 0 && b == 0) return c ? 0 : (x = y = 0, 1);
    g = extended_gcd(a, b, x, y);
    return (c % g == 0 ? (x *= c / g, y *= c / g, 1) : 0);
}
ll divdown(ll a, ll b) {
    return a / b - ((a ^ b) < 0 && a % b);
}

```

```

ll divup(ll a, ll b) {
    return a / b + ((a ^ b) >= 0 && a % b);
}
vector<pll> allsolution(ll a, ll b, ll c, ll lx, ll rx, ll
ly, ll ry) {
    // x = x0 + k*(b/g)    y = y0 - k*(a/g)
    ll x0, y0, g;
    if (!diophantine(a, b, c, x0, y0, g)) return {};
    vector<pll> ans;
    if (!g) {
        for (ll x = lx; x <= rx; x++)
            for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    if (!a) {
        ll y = c / b;
        if (!(ly <= y && y <= ry)) return {};
        for (ll x = lx; x <= rx; x++) ans.pb({x, y});
        return ans;
    }
    if (!b) {
        ll x = c / a;
        if (!(lx <= x && x <= rx)) return {};
        for (ll y = ly; y <= ry; y++) ans.pb({x, y});
        return ans;
    }
    ll dx = b / g, dy = a / g;
    ll kmin, kmax;
    if (dx > 0) kmin = divup(lx - x0, dx), kmax = divdown(rx
- x0, dx);
    else kmin = divup(rx - x0, dx), kmax = divdown(lx - x0,
dx);
    if (dy > 0) kmin = max(kmin, divup(y0 - ry, dy)), kmax =
min(kmax, divdown(y0 - ly, dy));
    else kmin = max(kmin, divup(y0 - ly, dy)), kmax =
min(kmax, divdown(y0 - ry, dy));
    for (ll k = kmin; k <= kmax; k++) ans.pb({x0 + k * dx,
y0 - k * dy});
    return ans;
}
bool positive_sol_minf(ll a, ll b, ll c, ll &x, ll &y, ll
cx, ll cy) {
    // f(x,y)= cx*x + cy*y
    ll g;
    if (!diophantine(a, b, c, x, y, g)) return 0;
    ll dx = b / g, dy = a / g;
    ll kmin = -INF, kmax = INF;
    dx > 0 ? kmin = max(kmin, divup(-x, dx)) : kmax =
min(kmax, divdown(-x, dx));
    dy < 0 ? kmin = max(kmin, divup(y, dy)) : kmax =
min(kmax, divdown(y, dy));
    if (kmin > kmax) return 0;
    ll df = cx * dx - cy * dy;
    ll k = df >= 0 ? kmin : kmax; // df<0 para max
    x += dx * k;

```

```

    y -= dy * k;
    return 1;
}

```

4.4 fft

```

typedef complex<double> cd;
vector<vector<cd>> w;
void calc(int log) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <= len) {
        w[l].resize(len / 2);
        cd wn(cos(2 * PI / len), sin(2 * PI / len));
        for (int j = 0; 2 * j < len; j++)
            w[l][j] = (j & 1 ? w[l - 1][j / 2] * wn : w[l -
1][j / 2]);
    }
}
void fft(vector<cd> &a, bool invert) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <= len)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                cd u = a[i + j], v = a[i + j + len / 2] *
w[l][j];
                a[i + j] = u + v;
                a[i + j + len / 2] = u - v;
            }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        for (int i = 0; i < n; i++) a[i] /= n;
    }
}
vector<ll> polymult(vector<ll> &a, vector<ll> &b) {
    int n = 1, log = 0;
    while (n < a.size() + b.size() - 1) n <= n, log++;
    calc(log);
    vector<cd> A(n);
    for (int i = 0; i < n; i++) A[i] = cd(i < a.size() ?
a[i] : 0, i < b.size() ? b[i] : 0);
    fft(A, 0);
    for (int i = 0; i < n; i++) A[i] *= A[i];
    fft(A, 1);
    vector<ll> ans(a.size() + b.size() - 1);
    for (int i = 0; i < ans.size(); i++) ans[i] =
round(A[i].imag() / 2);
    return ans;
}

```

```
// 2D-FFT
void fft2D(vector<vector<cd>> &a, bool invert) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) fft(a[i], invert);
    vector<cd> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        fft(b, invert);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}

vector<vector<ll>> convolution2D(vector<vector<ll>> &a,
vector<vector<ll>> &b) {
    int n = 1, logn = 0;
    while (n < a.size() + b.size() - 1) n <= 1, logn++;
    int m = 1, logm = 0;
    while (m < a[0].size() + b[0].size() - 1) m <= 1, logm++;
    calc(max(logn, logm));
    vector<vector<cd>> A(n, vector<cd>(m));
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++)
            A[i][j] = cd(i < a.size() && j < a[0].size() ?
a[i][j] : 0,
i < b.size() && j < b[0].size() ?
b[i][j] : 0);
    fft2D(A, 0);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) A[i][j] *= A[i][j];
    fft2D(A, 1);
    vector<vector<ll>> ans(a.size() + b.size() - 1,
vector<ll>(a[0].size() + b[0].size() - 1));
    for (int i = 0; i < ans.size(); i++)
        for (int j = 0; j < ans[0].size(); j++) ans[i][j] =
round(A[i][j].imag() / 2);
    return ans;
}
```

4.5 fwht bitwiseconvolution

```
//OR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j + h] +=
a[j];
}

void ifwht(vector<ll> &a) {
    int n = A.size();
    for (int s = n, h = s >> 1; s >= 2; s >= 1, h >= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j + h] -=
A[j];
}
```

```
vector<ll> OrConvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}

vector<ll> SubsetConvolution(vector<ll> a, vector<ll> b) {
    int n = 1, bit = 1;
    while (n < a.size() || n < b.size()) n <= 1, bit++;
    a.resize(n), b.resize(n);
    vector<vector<ll>> A(bit + 1, vector<ll>(n)), B(bit + 1,
vector<ll>(n));
    for (int i = 0; i < n; i++) {
        int f = __builtin_popcount(i);
        A[f][i] = a[i], B[f][i] = b[i];
    }
    for (int i = 0; i <= bit; i++) fwht(A[i]), fwht(B[i]);
    vector<ll> conv(n, 0);
    for (int k = 0; k < n; k++) {
        int f = __builtin_popcount(k);
        for (int i = 0; i <= f; i++) conv[k] += A[i][k] *
B[f - i][k];
    }
    ifwht(conv);
    return conv;
}

//AND-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) a[j] += a[j +
h];
}

void ifwht(vector<ll> &a) {
    int n = A.size();
    for (int s = n, h = n >> 1; s >= 2; s >= 1, h >= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) A[j] -= A[j +
h];
}

vector<ll> Andconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
```

```
//XOR-CONVOLUTION
void fwht(vector<ll> &a) {
    int n = a.size();
    for (int s = 2, h = 1; s <= n; s <= 1, h <= 1)
        for (int i = 0; i < n; i += s)
            for (int j = i; j < i + h; j++) {
                ll a1 = a[j + h], a0 = a[j];
                a[j + h] = a0 - a1, a[j] = a0 + a1;
            }
}

void ifwht(vector<ll> &a) {
    fwht(A);
    int n = A.size();
    for (int i = 0; i < n; i++) A[i] /= n;
}

vector<ll> Xorconvolution(vector<ll> a, vector<ll> b) {
    int n = 1;
    while (n < a.size() || n < b.size()) n <= 1;
    a.resize(n), b.resize(n);
    fwht(a), fwht(b);
    vector<ll> conv(n);
    for (int i = 0; i < n; i++) conv[i] = a[i] * b[i];
    ifwht(conv);
    return conv;
}
```

4.6 gauss

```
const int N = 3e5 + 9;
const double eps = 1e-9;
int Gauss(vector<vector<double>> a, vector<double> &ans) {
    int n = (int)a.size(), m = (int)a[0].size() - 1;
    vector<int> pos(m, -1);
    double det = 1;
    int rank = 0;
    for (int col = 0, row = 0; col < m && row < n; ++col) {
        int mx = row;
        for (int i = row; i < n; i++)
            if (fabs(a[i][col]) > fabs(a[mx][col])) mx = i;
        if (fabs(a[mx][col]) < eps) {
            det = 0;
            continue;
        }
        for (int i = col; i <= m; i++) swap(a[row][i], a[mx][
i]);
        if (row != mx) det = -det;
        det *= a[row][col];
        pos[col] = row;
        for (int i = 0; i < n; i++) {
            if (i != row && fabs(a[i][col]) > eps) {
                double c = a[i][col] / a[row][col];
                for (int j = col; j <= m; j++) a[i][j] -=
a[row][j] * c;
            }
        }
    }
}
```

```

    ++row;
    ++rank;
}
ans.assign(m, 0);
for (int i = 0; i < m; i++) {
    if (pos[i] != -1) ans[i] = a[pos[i]][m] / a[pos[i]]
[i];
}
for (int i = 0; i < n; i++) {
    double sum = 0;
    for (int j = 0; j < m; j++) sum += ans[j] * a[i][j];
    if (fabs(sum - a[i][m]) > eps) return -1; // no
solution
}
for (int i = 0; i < m; i++)
    if (pos[i] == -1) return 2; // infinte solutions
return 1; // unique solution
}

```

4.7 geo2d

```

struct point {
    ll x, y;
    point() {}
    point(ll _x, ll _y) : x(_x), y(_y) {}
    point operator+(const point &p) const {
        return {x + p.x, y + p.y};
    }
    point operator-(const point &p) const {
        return {x - p.x, y - p.y};
    }
    bool operator==(const point &p) const {
        return x == p.x && y == p.y;
    }
    bool operator<(const point &p) const {
        if (x == p.x) return y < p.y;
        return x < p.x;
    }
};
// Producto punto
long long dot(const point &a, const point &b) {
    return a.x * b.x + a.y * b.y;
}
// Producto cruzado (determinante)
long long cross(const point &a, const point &b) {
    return a.x * b.y - a.y * b.x;
}
// Cross entre vectores AB y AC
long long cross(const point &a, const point &b, const point
&c) {
    return (b.x - a.x) * (c.y - a.y) - (b.y - a.y) * (c.x -
a.x);
}
// El punto p está en el segmento ab?
bool onSegment(const point &a, const point &b, const point

```

```

&p) {
    if (cross(a, b, p) != 0) return false;
    return min(a.x, b.x) <= p.x && p.x <= max(a.x, b.x) &&
min(a.y, b.y) <= p.y &&
        p.y <= max(a.y, b.y);
}
// Interseccion de 2 segmentos
bool intersect(const point &p1, const point &q1, const point
&p2, const point &q2) {
    long long d1 = cross(p1, q1, p2);
    long long d2 = cross(p1, q1, q2);
    long long d3 = cross(p2, q2, p1);
    long long d4 = cross(p2, q2, q1);
    if ((d1 > 0 && d2 < 0 || d1 < 0 && d2 > 0) && (d3 > 0 &&
d4 < 0 || d3 < 0 && d4 > 0))
        return true;
    if (d1 == 0 && onSegment(p1, q1, p2)) return true;
    if (d2 == 0 && onSegment(p1, q1, q2)) return true;
    if (d3 == 0 && onSegment(p2, q2, p1)) return true;
    if (d4 == 0 && onSegment(p2, q2, q1)) return true;
    return false;
}
bool pointInPolygon(const vector<point> &poly, const point
&p) {
    bool inside = false;
    int n = poly.size();
    for (int i = 0, j = n - 1; i < n; j = i++) {
        const point &a = poly[i];
        const point &b = poly[j];
        if (onSegment(a, b, p)) return true;
        if ((a.y > p.y) != (b.y > p.y)) {
            long long x_intersection = (b.x - a.x) * (p.y -
a.y) - (b.y - a.y) * (p.x - a.x);
            long long y_diff = b.y - a.y;
            bool right = (x_intersection * y_diff) < 0;
            if (right) inside = !inside;
        }
    }
    return inside;
}

```

4.8 matrix

```

#define matrix array<array<int, N>, N>
matrix multiply(matrix &a, matrix &b) {
    matrix ans{};
    for (int i = 0; i < N; i++)
        for (int j = 0; j < N; j++)
            for (int k = 0; k < N; k++) ans[i][j] =
smod(ans[i][j], mult(a[i][k], b[k][j]));
    return ans;
}
matrix pow(matrix &a, ll e) {
    matrix ans{};
    for (int i = 0; i < N; i++) ans[i][i] = 1;
}

```

```

while (e) {
    if (e & 1) ans = multiply(ans, a);
    a = multiply(a, a);
    e >>= 1;
}
return ans;
}

```

4.9 millerrabin polardrho

```

ll mult(ll a, ll b, ll mod) {
    return (int128)a * b % mod;
}
ll qp(ll a, ll e, ll mod) {
    ll ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
bool compositetest(ll a, ll d, ll s, ll n) {
    ll x = qp(a, d, n);
    if (x == 1 || x == n - 1) return 0;
    for (ll i = 0; i < s - 1; i++) {
        x = mult(x, x, n);
        if (x == n - 1) return 0;
    }
    return 1;
}
bool esprimo(ll n) {
    if (n < 2) return 0;
    ll d = n - 1, s = 0;
    while ((d & 1) == 0) d >>= 1, s++;
    for (ll a : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31,
37}) {
        if (n == a) return 1;
        if (compositetest(a, d, s, n)) return 0;
    }
    return 1;
}
mt19937_64
rng(chrono::steady_clock::now().time_since_epoch().count());
ll random(ll a, ll b) {
    return uniform_int_distribution<ll>(a, b)(rng);
}
ll f(ll x, ll c, ll mod) {
    ll m = mult(x, x, mod);
    return m + c >= mod ? m + c - mod : m + c;
}
ll getfactor(ll n) {
    if (n % 2 == 0) return 2;
    if (esprimo(n)) return n;
    while (true) {

```



```

ll y = random(1, n - 1);
ll c = random(1, n - 1);
ll m = 64, g = 1, r = 1, q = 1;
ll x, ys;
while (g == 1) {
    x = y;
    for (ll i = 0; i < r; i++) y = f(y, c, n);
    ll k = 0;
    while (k < r && g == 1) {
        ys = y;
        ll lim = min(m, r - k);
        for (ll i = 0; i < lim; i++) {
            y = f(y, c, n);
            ll diff = x > y ? x - y : y - x;
            q = mult(q, diff, n);
        }
        g = __gcd(q, n);
        k += m;
    }
    r <<= 1;
}
if (g == n) {
    do {
        ys = f(ys, c, n);
        ll diff = x > ys ? x - ys : ys - x;
        g = __gcd(diff, n);
    } while (g == 1);
}
if (g > 1 && g < n) return g;
}
}
vector<pll> fp(ll n) {
    stack<ll> st;
    st.push(n);
    vector<pll> ans;
    map<ll, ll> cont;
    while (!st.empty()) {
        ll x = st.top();
        st.pop();
        if (x == 1) continue;
        if (esprimo(x)) cont[x]++;
        else {
            ll f = getfactor(x);
            st.push(f), st.push(x / f);
        }
    }
    for (auto [x, f] : cont) ans.pb({x, f});
    return ans;
}

```

4.10 ntt

```

const ll MOD = 998244353;
vector<vector<int>> w;

```

```

int pow2, root;
int smod(int a, int b, int mod) {return (a + b >= mod ? a + b - mod : a + b);}
int rmod(int a, int b, int mod) {return (a - b < 0 ? a - b + mod : a - b);}
int mult(int a, int b, int mod) {return (ll)a * b % mod;}
int qp(int a, int e, int mod) {
    int ans = 1;
    while (e) {
        if (e & 1) ans = mult(ans, a, mod);
        a = mult(a, a, mod);
        e >>= 1;
    }
    return ans;
}
void calc(int log, int mod) {
    w.resize(log + 1);
    w[0].resize(1, 1);
    for (int l = 1, len = 2; l <= log; l++, len <<= 1) {
        w[l].resize(len / 2);
        int wn = qp(root, pow2 / len, mod);
        for (int j = 0; 2 * j < len; j++) {
            if (j & 1) w[l][j] = mult(w[l - 1][j / 2], wn, mod);
            else w[l][j] = w[l - 1][j / 2];
        }
    }
}
void findroot(int mod) {
    pow2 = 1;
    int k = mod - 1;
    while (!(k & 1)) pow2 <<= 1, k >>= 1;
    root = 1;
    while (qp(root, pow2, mod) != 1 || qp(root, pow2 / 2, mod) == 1) root++;
}
void ntt(vector<int> &a, bool invert, int mod) {
    int n = a.size();
    for (int i = 1, j = 0; i < n; i++) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int l = 1, len = 2; len <= n; l++, len <<= 1)
        for (int i = 0; i < n; i += len)
            for (int j = 0; 2 * j < len; j++) {
                int u = a[i + j], v = mult(a[i + j + len / 2], w[l][j], mod);
                a[i + j] = smod(u, v, mod);
                a[i + j + len / 2] = rmod(u, v, mod);
            }
    if (invert) {
        reverse(a.begin() + 1, a.end());
        int invn = qp(n, mod - 2, mod);
    }
}

```

```

        for (int i = 0; i < n; i++) a[i] = mult(a[i], invn, mod);
    }
}
vector<int> convolution(vector<int> a, vector<int> b, int mod = MOD) {
    int n = 1, log = 0, rn = a.size() + b.size() - 1;
    while (n < a.size() + b.size() - 1) n <<= 1, log++;
    findroot(mod);
    calc(log, mod);
    a.resize(n), b.resize(n);
    ntt(a, 0, mod), ntt(b, 0, mod);
    for (int i = 0; i < n; i++) a[i] = mult(a[i], b[i], mod);
    ntt(a, 1, mod);
    a.resize(rn);
    return a;
}
// 2D-FFT
void ntt2D(vector<vector<int>> &a, bool invert, int mod) {
    int n = a.size(), m = a[0].size();
    for (int i = 0; i < n; i++) ntt(a[i], invert, mod);
    vector<int> b(n);
    for (int j = 0; j < m; j++) {
        for (int i = 0; i < n; i++) b[i] = a[i][j];
        ntt(b, invert, mod);
        for (int i = 0; i < n; i++) a[i][j] = b[i];
    }
}
vector<vector<int>> convolution2D(vector<vector<int>> a, vector<vector<int>> b, int mod = MOD) {
    int n = 1, logn = 0, rn = a.size() + b.size() - 1;
    while (n < rn) n <<= 1, logn++;
    int m = 1, logm = 0, rm = a[0].size() + b[0].size() - 1;
    while (m < rm) m <<= 1, logm++;
    findroot(mod);
    calc(max(logn, logm), mod);
    a.resize(n), b.resize(n);
    for (int i = 0; i < n; i++) a[i].resize(m);
    b[i].resize(m);
    ntt2D(a, 0, mod), ntt2D(b, 0, mod);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < m; j++) a[i][j] = mult(a[i][j], b[i][j], mod);
    ntt2D(a, 1, mod);
    a.resize(rn);
    for (int i = 0; i < rn; i++) a[i].resize(rm);
    return a;
}

```

4.11 primitive root

```

int primitive_root(int p) {
    int n = p - 1; // n=phi(p) -> n=p-1 si p es primo
}

```



```
vector<int> fp;
for (int i = 2; i * i <= n; i++)
    if (n % i == 0) {
        fp.pb(i);
        while (n % i == 0) n /= i;
    }
if (n > 1) fp.pb(n);
n = p - 1;
bool ok = 0;
for (int g = 2; g <= n; g++) {
    bool ok = 1;
    for (int x : fp)
        if (qp(g, n / x, p) == 1) {
            ok = 0;
            break;
        }
    if (ok) return g;
}
return 0;
```

4.12 random

```
static mt19937 rng(chrono::steady_clock::now()
    .time_since_epoch().count());
#define rnd(a,b) (uniform_int_distribution<ll>(a,b)(rng))
```

4.13 theorems

4.13.1 Combinatoria

4.13.1.1 General

- $\sum_{k=0}^n \binom{n-k}{k} = \text{Fib}_{n+1}$
- $\binom{n}{k} = \binom{n}{n-k}$
- $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$
- $k \binom{n}{k} = n \binom{n-1}{k-1}$
- $\sum_{i=0}^n \binom{n}{i} = 2^n$

Identidad de Vandermonde:

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

Identidad del Palo de Hockey (Hockey-Stick):

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

4.13.1.2 Triángulo de Pascal

- En una fila p donde p es un número primo, todos los términos en esa fila excepto los unos son múltiplos de p .
- Teorema de Kummer:** Para enteros $n \geq m \geq 0$ y un número primo p , la mayor potencia de p que divide a $\binom{n}{m}$ es igual al número de acarreo cuando m se suma a $n - m$ en base p .

4.13.1.3 Números de Catalan

- Fórmula general: $C_n = \frac{1}{n+1} \binom{2n}{n}$
- Recurrencia: $C_0 = 1, C_1 = 1$ y $C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}$

Representa, entre otras cosas, el número de secuencias de paréntesis balanceadas o triangulaciones de un polígono convexo.

4.13.1.4 Números de Stirling (Primera y Segunda Especie)

- Primera especie** (número de permutaciones con k ciclos disjuntos):

$$S(n, k) = (n-1) \cdot S(n-1, k) + S(n-1, k-1)$$
- Segunda especie** (maneras de particionar un conjunto de n objetos en k subconjuntos no vacíos):

$$S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$$

4.13.1.5 Números de Bell

Cuenta el número total de particiones posibles de un conjunto.

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} \cdot B_k$$

4.13.2 Matemáticas

4.13.2.1 Fórmulas Generales

- $a^k - b^k = (a-b) \cdot (a^{k-1}b^0 + a^{k-2}b^1 + \dots + a^0b^{k-1})$
- $|a-b| + |b-c| + |c-a| = 2(\max(a, b, c) - \min(a, b, c))$

Identidad de Lagrange:

$$\left(\sum_{k=1}^n a_k^2 \right) \left(\sum_{k=1}^n b_k^2 \right) - \left(\sum_{k=1}^n a_k b_k \right)^2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n (a_i b_j - a_j b_i)^2$$

4.13.2.2 Fórmulas de Vieta

Para un polinomio de grado n : $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ con raíces $r_1, r_2, \dots, r_n$:

- $r_1 + r_2 + \dots + r_n = -\frac{a_{n-1}}{a_n}$
- $r_1 r_2 \dots r_n = (-1)^n \frac{a_0}{a_n}$

4.13.2.3 Sucesión de Fibonacci

- $F_n = \sum_{k=0}^{\lfloor \frac{n-1}{2} \rfloor} \binom{n-k-1}{k}$
- Fórmula de Binet:** $F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right)$
- $\sum_{i=1}^n F_i = F_{n+2} - 1$
- $\gcd(F_m, F_n) = F_{\gcd(m, n)}$

4.13.2.4 Ternas Pitagóricas

Fórmula de Euclides para generar ternas donde $a^2 + b^2 = c^2$: Para $m > n > 0$:

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2$$

4.13.3 Teoría de Números

4.13.3.1 Función Totiente de Euler (φ)

- $\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$ donde p son factores primos distintos.
- Si a y b son coprimos, entonces $\varphi(a \cdot b) = \varphi(a) \cdot \varphi(b)$

$$\sum_{d|n} \varphi(d) = n$$

4.13.3.2 Funci3n e Inversi3n de M3bius

$\mu(n)$ toma valores en $\{-1, 0, 1\}$:

- 1 si es libre de cuadrados con cantidad par de factores primos.
- -1 si es libre de cuadrados con cantidad impar de factores primos.
- 0 si contiene alg3n factor primo al cuadrado.

Teorema de inversi3n de M3bius: Si

$g(n) = \sum_{d|n} f(d)$, entonces:

$$f(n) = \sum_{d|n} \mu(d)g\left(\frac{n}{d}\right)$$

4.13.3.3 MCD (GCD) y MCM (LCM)

- $\gcd(a, b) \cdot \text{lcm}(a, b) = |a \cdot b|$
- $\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$
- $\sum_{k=1}^n \gcd(k, n) = \sum_{d|n} d \cdot \varphi\left(\frac{n}{d}\right)$

4.13.4 Operaciones de Bits (Bitwise) y Miscel3nea

- $a + b = (a \oplus b) + 2(a \text{ AND } b)$
- $a + b = (a | b) + (a \text{ AND } b)$
- El k -3simo bit est3 encendido en x si y solo si $x \bmod 2^k \geq 2^{k-1}$.
- $1 \oplus 2 \oplus 3 \oplus \dots \oplus (4k - 1) = 0$ para todo $k \geq 0$.

4.13.4.1 Teorema de Erd3s3s Gallai

Una secuencia descendente de grados $d_1 \geq d_2 \geq \dots \geq d_n$ puede representar un grafo simple finito si la suma total de grados es par y:

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min(d_i, k)$$

para cada $1 \leq k \leq n$

4.14 xor basis

```
template <typename T, int LOG>
struct Basis {
    T basis[LOG] = {0};
    int dim = 0;
    bool insert(T x) { // Formar3 parte del Basis?
        for (int i = LOG - 1; i >= 0; i--)
            if ((x >> i) & 1) {
                if (basis[i] x ^= basis[i];
                else {
                    basis[i] = x, dim++;
                    return true;
                }
            }
        return false;
    }
    T size() {
        return ((T)1 << dim);
    }
    bool find(T x) { // 3x 3^ span{x1,x2...}?
        for (int i = LOG - 1; i >= 0; i--) x = min(x, x ^
basis[i]);
        return x == 0;
    }
    T max_xor(T x = 0) { // max (x^a / a 3^ span{x1,x2...})
        for (int i = LOG - 1; i >= 0; i--) x = max(x, x ^
basis[i]);
        return x;
    }
    T kth(T k) { // The k_smaller of span{x1,x2...}
        if (k < 1 || k > ((T)1 << dim)) return -1;
        T x = 0, cont = ((T)1 << dim);
        for (int i = LOG - 1; i >= 0; i--)
            if (basis[i]) {
                if (k > (cont >> 1)) {
                    if (!((x >> i) & 1)) x ^= basis[i];
                    k -= (cont >> 1);
                } else if ((x >> i) & 1) x ^= basis[i];
                cont >>= 1;
            }
        return x;
    }
}
```

5 Strings

5.1 aho corasick

```
struct Aho_Corasick{
    int t[N][K], cur;
    int pa[N], go[N][K], lnk[N];
    queue<int> qu;

    int insert(string &s){
        int u = 0;
```

```
        for (char &c : s){
            int k = c-'a';
            if (t[u][k] == 0)
                t[u][k] = ++cur, pa[cur] = u;
            u = t[u][k];
        }
        return u;
    }

    void bfs(){
        fill(lnk+1, lnk+cur+1, -1);
        for (qu.push(0); !qu.empty(); ){
            int u = qu.front(); qu.pop();
            for (int k=0; k<K; k++){
                int v = t[u][k];
                if (v == 0) go[u][k] = go[lnk[u]][k];
                else{
                    lnk[v] = u == 0 ? 0 : go[lnk[u]][k];
                    go[u][k] = t[u][k];
                    qu.push(v);
                }
            }
        }
    }
} aho; // by NekoRolly
```

5.2 hashing

```
struct Hashing{
    const int mod = 1e9+9;
    const int p = 367, q = 467;
    int ep[N], eq[N], prp[N], prq[N];

    int P(int a, int b){ return 1ll*a*b%mod; }
    int S(int a, int b){ return (a+b)%mod; }

    void build(string &s){
        int n = s.size();
        ep[0] = eq[0] = 1;
        prp[0] = prq[0] = 0;
        for (int i=1; i<=n; i++){
            prp[i] = S(P(prp[i-1], p), s[i-1]);
            prq[i] = S(P(prq[i-1], q), s[i-1]);
            ep[i] = P(ep[i-1], p);
            eq[i] = P(eq[i-1], q);
        }
    }

    ll substr(int i, int len){ i++;
        ll hp = S(prp[i+len-1], mod-P(prp[i-1], ep[len]));
        ll hq = S(prq[i+len-1], mod-P(prq[i-1], eq[len]));
        return hp*mod + hq;
    }
} H1, H2; // by NekoRolly
```

5.3 manacher

```
// d1[i] odd number of palindromes centered at i
// d2[i] even number of palindromes centered at (i, i + 1)
void manacher(string &s, vector<int> &d1, vector<int> &d2) {
    int n = s.size();
    d1 = vector<int>(n);
    d2 = vector<int>(n, 0);
    int l = -1, r = -1;
    for (int i = 0; i < n; i++) {
        d1[i] = (i > r) ? 1 : min(d1[l + r - i], r - i + 1);
        while (i - d1[i] >= 0 && i + d1[i] < n && s[i - d1[i]] == s[i + d1[i]]) {
            d1[i]++;
        }
        if (i + d1[i] - 1 > r) r = i + d1[i] - 1, l = i - d1[i] - 1;
    }
    l = -1, r = -1;
    for (int i = 0; i < n - 1; i++) {
        d2[i] = (i >= r) ? 0 : min(d2[l + r - i - 1], r - i);
        while (i - d2[i] >= 0 && i + d2[i] + 1 < n && s[i - d2[i]] == s[i + d2[i] + 1]) {
            d2[i]++;
        }
        if (i + d2[i] > r && d2[i] > 0) r = i + d2[i], l = i - d2[i] + 1;
    }
}
```

5.4 prefix function

```
void prefix_function(string &s, int pf[]) {
    int n = s.size(); pf[0] = 0;
    for (int i = 1; i < n; i++) {
        int j = pf[i - 1];
        while (j > 0 && s[j] != s[i]) j = pf[j - 1];
        pf[i] = j + (s[i] == s[j]);
    }
} // by NekoRolly
```

5.5 suffix array

```
enum Comparison { LESS, EQUAL, GREATER };

Comparison getComparison(int a, int b) {
    if (a < b) return Comparison::LESS;
    if (a > b) return Comparison::GREATER;
    return Comparison::EQUAL;
}

const int ALPH = 256;
struct SuffixArray {
    vector<int> suffixArray;
```

```
vector<int> lcp;
// lcp[i] = largest common preffix of sa[i] and sa[i + 1] (in sorted list)
vector<int> pos;

vector<int> sortCyclic(string &s) { // O(n log n)
    int n = s.size();
    vector<int> p(n), c(n), cnt(max(ALPH, n), 0);
    // p[] : sorted array with the starting index of substrings of length 2^k
    // c[] : equivalence class
    // counting sort
    for (int i = 0; i < n; i++) cnt[s[i]]++;
    for (int i = 1; i < ALPH; i++) cnt[i] += cnt[i - 1];
    for (int i = 0; i < n; i++) p[--cnt[s[i]]] = i;

    c[p[0]] = 0;
    int classes = 1;
    for (int i = 1; i < n; i++) {
        if (s[p[i]] != s[p[i - 1]]) classes++;
        c[p[i]] = classes - 1;
    }
    // radix sort
    vector<int> pNew(n), cNew(n);
    for (int k = 0; (1 << k) < n; k++) {
        // sort pNew by the second substring
        for (int i = 0; i < n; i++) {
            pNew[i] = p[i] - (1 << k);
            if (pNew[i] < 0) pNew[i] += n;
            cnt[i] = 0;
        }
        // counting sort in the first substring
        for (int i = 0; i < n; i++) cnt[c[pNew[i]]]++;
        for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
        for (int i = n - 1; i >= 0; i--) {
            p[--cnt[c[pNew[i]]]] = pNew[i];
        }
        cNew[p[0]] = 0;
        classes = 1;
        for (int i = 1; i < n; i++) {
            int x = p[i] + (1 << k);
            if (x >= n) x -= n;
            int y = p[i - 1] + (1 << k);
            if (y >= n) y -= n;
            pair<int, int> cur = {c[p[i]], c[x]};
            pair<int, int> prev = {c[p[i - 1]], c[y]};
            if (cur != prev) classes++;
            cNew[p[i]] = classes - 1;
        }
        c = cNew;
    }
    return p;
}
```

```
void buildLCP(string &s) { // O(n)
    int n = s.size();
    lcp = vector<int>(n - 1, 0);
    pos = vector<int>(n);
    for (int i = 0; i < n; i++) pos[suffixArray[i]] = i;
    int k = 0;
    for (int r = 0; r < n; r++) {
        if (pos[r] == 0) continue;
        int l = suffixArray[pos[r] - 1];
        while (s[l + k] == s[r + k]) k++;
        lcp[pos[r] - 1] = k;
        k = max(0, k - 1);
    }

    // Suffix array: contain starting indexes of the all the suffixes (sorted)
    SuffixArray(string &s) { // O(n log n)
        char minChar = 'a' - 1;
        s += minChar;
        suffixArray = sortCyclic(s);
        suffixArray.erase(suffixArray.begin());
        s.pop_back();
        buildLCP(s);
    }

    // return the lcp of the suffixes s[l...n - 1] and s[r...n - 1]
    int getLCP(int l, int r) {
        assert(l != r);
        int minPos = min(pos[l], pos[r]);
        int maxPos = max(pos[l], pos[r]);
        return min(lcp[minPos], ..., lcp[maxPos - 1])
    }

    /*Comparison compare(int l1, int r1, int l2, int r2) {
        // compare two substrings [l1, r1], [l2, r2].
        int curlCP = (l1 == l2) ? suffixArray.size() :
        getLCP(l1, l2);
        if (r1 < l1 + curlCP) {
            if (r2 < l2 + curlCP) {
                int sz1 = r1 - l1 + 1;
                int sz2 = r2 - l2 + 1;
                return getComparison(sz1, sz2);
            } else return Comparison::LESS;
        } else if (r2 < l2 + curlCP) return
        Comparison::GREATER;
        return getComparison(pos[l1], pos[l2]);
    }
};
```

5.6 z function

```
void z_function(string &s, int zf[]) {
    int n = s.size();
```

```
    for (int i=1,l=0,r=0; i<n; i++){  
        if (i < r) zf[i] = min(zf[i-l], r-i);  
        while (i+zf[i] < n && s[i+zf[i]] == s[zf[i]]) zf[i]+  
+;        if (i+zf[i] > r) l = i, r = i+zf[i];  
    }  
} // by NekoRolly
```